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ASR Error Correction in Low-Resource Burmese with Alignment-Enhanced Transformers using Phonetic Features

Ye Bhone Lin[1+], Thura Aung[12+], Ye Kyaw Thu[13*], Thazin Myint Oo[1]

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2. Dataset Preparation
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1. Introduction

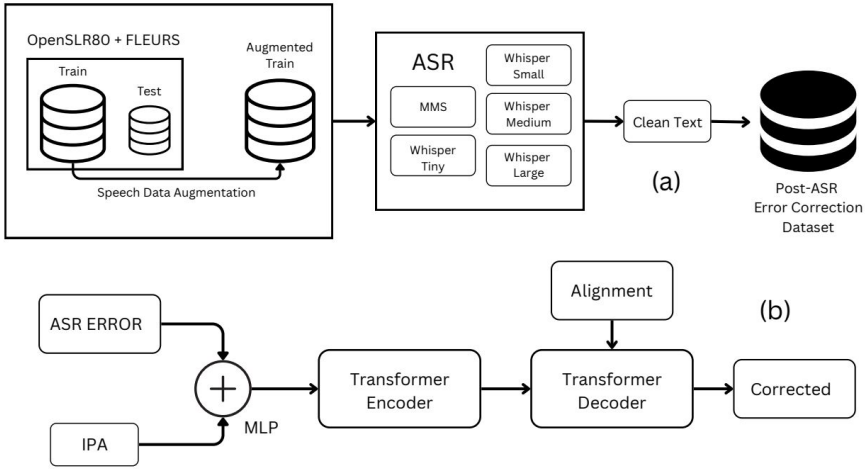

Fig. 1: (a) Post-ASR Dataset Preparation for ASR Error Correction (AEC) Training (b) Integration of Phonetic Features and Alignment in AEC.

- Automatic Speech Recognition (ASR) systems transcribe spoken language into text.
- However, they often produce **errors**, especially in **low-resource languages** like Burmese, due to limited training data.
- These transcription errors degrade performance in downstream tasks such as translation or information retrieval.

1. Introduction (Cont'd)

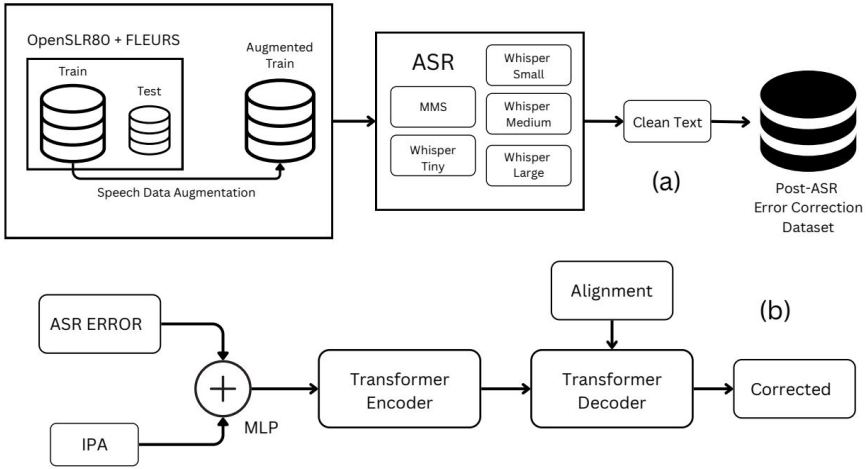

Fig. 1: (a) Post-ASR Dataset Preparation for ASR Error Correction (AEC) Training (b) Integration of Phonetic Features and Alignment in AEC.

- Recent Scientists, Fine-tuned BART model to correct ASR phonetic and spelling errors, improving WER on accented speech.
- The study proposes an **Alignment-Enhanced Transformer** that integrates **phonetic (IPA)** and **alignment** features for **ASR Error Correction (AEC)**.
- This is the **first work** to address Burmese ASR error correction with deep alignment and phonetic integration.

2. Dataset Preparation

Dataset	Train Hr.	Test Hr.	MOSNet
OpenSLR80	3.70	0.42	4.06
FLEURS	15.95	1.64	4.14

TABLE I: Speech Data Information

MOSNet (Mean Opinion Score)

- Evaluate the quality of Voice Conversion (VC) and TTS
- A Higher Score is Better
- We adopt the CNN-BLSTM MOS prediction model proposed by Lo et al. (2019), commonly known as MOSNet.

2. Dataset Preparation (Cont'd)

Data were **augmented (10%)** with:

- Pitch, shift, speed perturbation, background noise,
- Vocal Tract Length Perturbation (VTLP), and
- Time and frequency masking.

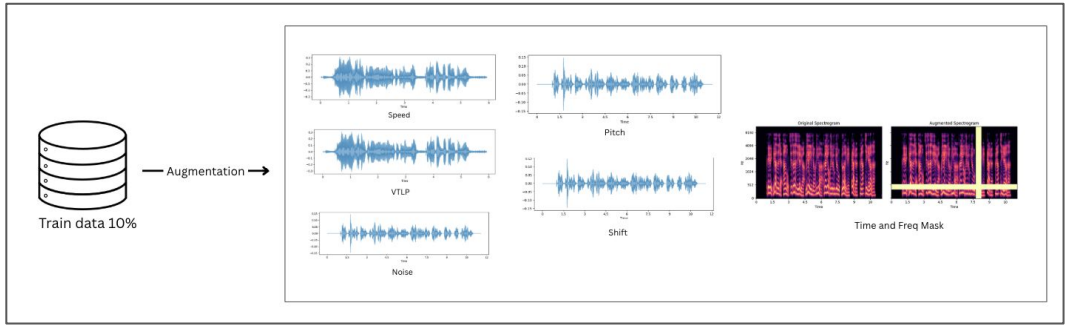

Fig. 2: Augmentation Pipeline

Data were segmented into syllables using **myWord**, ensuring linguistic consistency between ASR output and target text.

2. Dataset Preparation (Cont'd)

Model	Parameters	Languages
MMS-1B ASR	~1B	1,100+
Whisper Tiny*	39M	99
Whisper Small*	244M	99
Whisper Medium*	769M	99
Whisper Large*	1.55B	99

TABLE III: Pretrained ASR model information: model, number of parameters, and language coverage. * Whisper models were fine tuned for Burmese

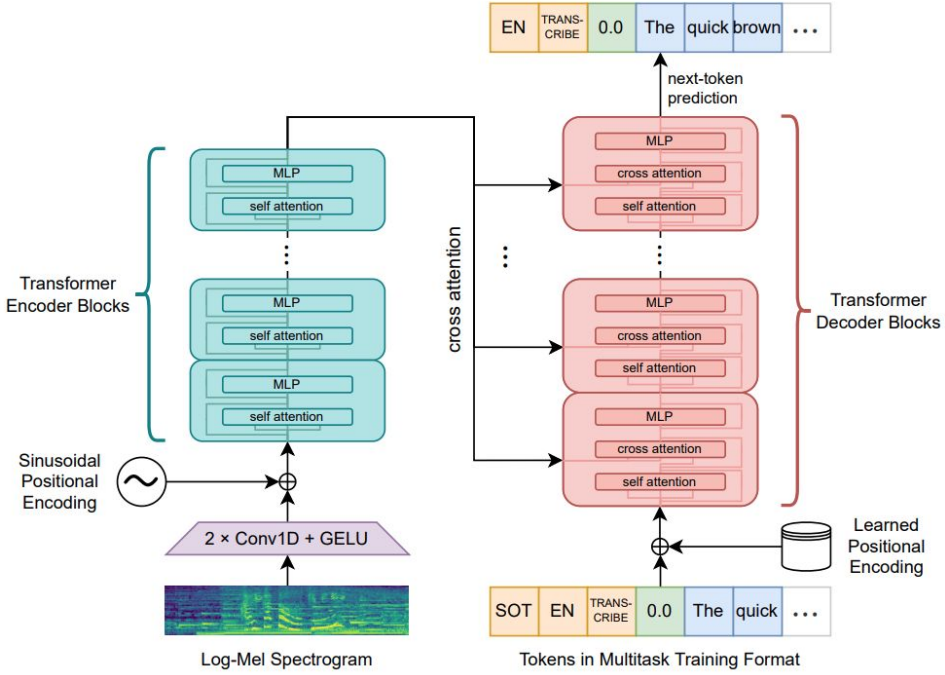

Fig. 3: Whisper Architecture

2. Dataset Preparation (Cont'd)

- Fine-tuned **Whisper models** — *Tiny, Small, Medium,* and *Large* on Burmese speech data to generate ASR outputs.
- Applied **Zero-shot learning** using **Meta MMS** to extend coverage for unseen speech patterns.
- Combined ASR hypotheses and ground truth transcriptions to build a **parallel dataset** for **ASR Error Correction (AEC)**.
- The resulting dataset contains aligned **error–correction pairs**, serving as input for AEC model training.

3. Materials

Framework: OpenNMT (Transformer sequence-to-sequence model).

ASR Sources: Whisper (Tiny, Medium, Large) and Meta MMS.

Feature Extraction Tools:

- **Phonetic (IPA):** myG2P v2.0 (CRF-based grapheme-to-phoneme conversion).
- **Alignment:** fast-align tool (for bilingual alignment scoring).

Hardware: NVIDIA GPU with 24GB VRAM, PyTorch backend.

No.	Words	Standard	Correct
1	သတင်းစာ (newspaper)	tha. tin: sa	dha- din: za
2	ပိုင်းခြေ (denominator)	pain: chei	pain: gyei
3	ပုလဲ (pearl)	pu. le:	pa- le:
4	ပညာ (knowledge)	pa. nja	pjin nja
5	မင်ကြောင် (tatoo)	min kyaun	mhin gyaun

TABLE IV: Examples of contextually dependent pronunciations of some Myanmar words

4. Experimental Setup

Model Architecture: 4-layer Transformer encoder-decoder (512 hidden units, 8 heads).

Optimization: Adam optimizer with Noam learning rate scheduler.

Batch size: 4096 tokens.

Metrics:

- **Word Error Rate (WER)** — measures overall transcription accuracy.
- **Character F-score (chrF++)** — measures character-level fluency and correctness.

Baseline Models: Whisper Tiny, Medium, Large and MMS ASR outputs.

5. Results and Discussion

Original Data				+ Augmentation Data			
ASR Model	**Feature**	**WER**	**chrF++**	**ASR Model**	**Feature**	**WER**	**chrF++**
MMS	No AEC (Baseline)	42.27	0.6126	MMS	No AEC (Baseline)	42.27	0.6126
	+ AEC	30.70	0.6461		+ AEC	**33.76**	**0.6670**
	+ AEC + IPA	31.40	0.6923		+ AEC + IPA	38.85	0.6331
	+ AEC + Align	**30.21**	**0.6940**		+ AEC + Align	34.07	0.6646
	+ AEC + IPA + Align	35.03	0.6596		+ AEC + IPA + Align	36.57	0.6605
Whisper Tiny	No AEC (Baseline)	55.07	0.5483	Whisper Tiny	No AEC (Baseline)	55.07	0.5483
	+ AEC	**43.79**	**0.5871**		+ AEC	**48.28**	**0.6245**
	+ AEC + IPA	45.21	0.5797		+ AEC + IPA	52.00	0.5361
	+ AEC + Align	44.17	0.5868		+ AEC + Align	48.43	0.5546
	+ AEC + IPA + Align	45.48	0.5463		+ AEC + IPA + Align	51.37	0.5816
Whisper Small	No AEC (Baseline)	37.25	0.6534	Whisper Small	No AEC (Baseline)	37.25	0.6534
	+ AEC	**32.57**	0.6745		+ AEC	**36.22**	0.6302
	+ AEC + IPA	33.58	0.6709		+ AEC + IPA	38.57	0.6329
	+ AEC + Align	32.71	**0.6763**		+ AEC + Align	36.27	0.6484
	+ AEC + IPA + Align	33.71	0.6338		+ AEC + IPA + Align	38.23	**0.6727**
Whisper Medium	No AEC (Baseline)	72.18	0.5425	Whisper Medium	No AEC (Baseline)	72.18	0.5425
	+ AEC	**41.88**	0.6148		+ AEC	**46.01**	0.5884
	+ AEC + IPA	43.50	0.6030		+ AEC + IPA	49.41	0.5629
	+ AEC + Align	42.05	**0.6169**		+ AEC + Align	46.51	0.5844
	+ AEC + IPA + Align	43.21	0.5616		+ AEC + IPA + Align	49.86	**0.6041**
Whisper Large	No AEC (Baseline)	51.04	0.5752	Whisper Large	No AEC (Baseline)	51.04	0.5752
	+ AEC	**36.38**	**0.6666**		+ AEC	39.87	**0.6225**
	+ AEC + IPA	38.08	0.6425		+ AEC + IPA	43.21	0.6028
	+ AEC + Align	36.47	0.6527		+ AEC + Align	**40.17**	0.6211
	+ AEC + IPA + Align	41.66	0.6205		+ AEC + IPA + Align	41.94	0.6141

TABLE V: Comparison of ASR Error Correction models trained with Different Features on both Original and Augmented Data Across Different ASR systems (Best results per ASR model in Bold)

5. Results and Discussion (Cont'd)

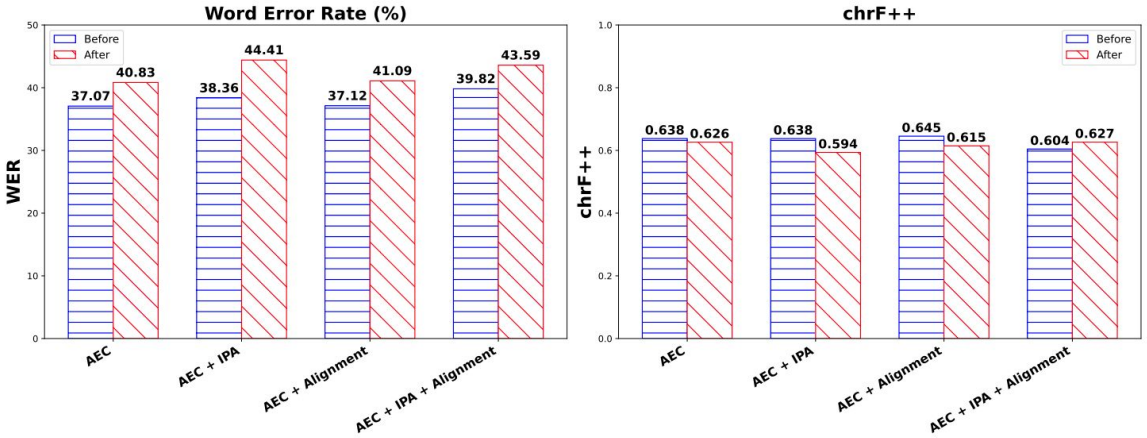

Fig. 4: Average WER and chrF++ Score Across Different AEC Approaches Before and After Data Augmentation.

- The **Word Error Rate (WER)** increased slightly after data augmentation across all AEC configurations.
- The **chrF++ scores** showed minor variations, with the *AEC + IPA + Alignment* approach showing a small improvement.
- Results suggest that data augmentation introduced some noise at the word level but helped maintain or improve **character-level consistency** in some models.
- The results validate that **linguistic-informed AEC models** can meaningfully enhance low-resource ASR systems without retraining acoustic models.

6. Error Analysis

1. ASR Error Analysis

All ASR models capture the general sentence structure but differ in accuracy at the **word and syllable level**.

ASR Models	Sentence
Groundtruth	စိတ်|sei' ဝင်|win| စား|za: ဖွယ်|bwe ကောင်း|gaun: သော|tho: ရွာ|jwa သို့|dhou. အေး|ei: အေး|ei: လူ|lu လူ|lu ဖြင့်|hpjin. နာ|na ရီ|ji ဝက်|we' ခန့်|khan. လမ်း|lan: လျှောက်|shau' သွား|dhwa: ရ|ra— သည်|dhe
MMS	အ|a- စတ်|ho ဝင်|win စာ|za ဘဲ|be: ကောင်|gaun တော်|do ရွာ|jwa သို့|dhou. a|x a|x လူ|lu ဖြ|bu. င့်|bu. နိုင်|bu. ဝက်|we' ခန့်|khan. လမ်း|lan: လျှောက်|shau' သွား|dhwa: ရ|ra- သည်|dhe
Whisper Tiny	ဆိုက်|hsai' ဝင်|win ကြ|kya- ပဲ|pe: ကောင်း|kaun: လော်|lo ရာ|ja သို့|dhou. Aိတ်|x အေး|ei: ဘီ|bi လူ|lu ဖြင့်|hpjin. မ|ma- ဟုတ်|hou' ပွဲ|pwe: ခံ|gan နဲ့|ne: ရှောက်|shau' ဘွား|hpwa: ရာ|ja သည်|dhi
Whisper Small	စိတ်|sei' ဝင်|win စာ|sa- ဘက်|be' ကောင်း|kaun: လောင်|laun ရွာ|jwa သို့|dhou. အေး|ei: အေး|ei: လူ|lu ဖြင့်|hpjin. နာ|na ရီ|ji ဝက်|we' ခန့်|khan. လမ်း|lan: လျှောက်|shau' သွား|dhwa: ရ|ra- သည်|dhe
Whisper Medium	စိတ်|sei' ဝင်|win စား|za: ဝ|we: ကောင်း|gaun: လော်|lo ရာ|ja သူ:|htei အင်|in အီ|i ယူ|ju ဖြင့်|hpjin. နိုင်|nain ဝက်|we' ခံ|gan နှင်:|nhin: ရှောက်|shau' သွား|thwa: ရ|ja. လေ|lei
Whisper Large	စိတ်|sei' ဝင်|win စာ|sa ပဲ|be: ကောင်:|gaun: လော်|lo ရ|ra' သူ|dhu ကျင်|gyin အေ|ei ဌ||hta ဖြင့်|hpjin. နိုင်|nain ဝက်|we' ခံ|khan နှ|nhan. လျှောက်|shau' သွား|thwa: ရ|ja. လေ|lei

TABLE VI: Error Comparison of Different ASR Output (Errors are colored in Red).

- **Whisper Medium and Large** produce fluent sentences but sometimes **hallucinate or substitute** words that sound similar.
- **Whisper Tiny and MMS** has **unstable decoding**, producing non-standard or repeated syllables such as “အ|a- စတ်|ho” or “a|x a|x”.

6. Error Analysis (Cont'd)

2. AEC Error Analysis

Applying AEC **reduces major lexical and syllable errors**, recovering correct forms closer to the ground truth.

AEC Approaches	Sentence
Correct	ရေ|jei စီး|zi: ကြောင်း|gyaun: မှ|mha. ထွက်|htwe' လာ|la သည်|thi နှင့်|ne. ရေ|jei ကူး|gu: ခြင်း|gyin: သည်|dhe ပုံ|boun မှန်|mhan အား|a: ဖြင့်|hpjin. မ|ma-ခက်|khe' ခဲ|ge: တော့|do. ပါ|ba
Whisper Tiny	ယ|ja- စီး|si: ချောင်း|chaun: မှ|mha. ထွက်|htwe' လာ|la သည်|dhe နှင့်|ne. ရေ|jei ကို|kou ချင်|gyo သည်|thi ပုံ|boun မှန်|mhan အား|a: ပြင်|bjin မ|ma-ချက်|che' ခဲ့|shoun: တော့|to. ပါ|ba
+ AEC	ရေ|jei စီး|zi: ကြောင်း|gyaun: မှာ|mha ထွက်|dwe' လာ|la သည်|dhe နှင့်|ne. ရေ|jei ကူး|gu: ခြင်း|gyin: သည်|dhe ပုံ|boun မှန်|mhan အား|a: ဖြင့်|hpjin. မ|ma-ခက်|khe' ခဲ|ge: သော|tho: ပါ|ba
+ AEC + IPA	ရေ|jei စီး|zi: ကြောင်း|gyaun: မှ|mha. ထွက်|htwe' လာ|la သည်|thi နှင့်|ne. ရေ|jei ကူး|gu: ခြင်း|gyin: သည်|dhe ပုံ|boun မှန်|mhan အား|a: ဖြင့်|hpjin. မ|ma-ခက်|khe' ခဲ|ge: တော့|do. ပါ|ba
+ AEC + Align	ရေ|jei စီး|zi: ကြောင်း|gyaun: မှာ|mha ထွက်|dwe' လာ|la ပြီ|pji နှင့်|ne. ရေ|jei ကူး|gu: ခြင်း|gyin: သည်|dhe ပုံ|boun မှန်|mhan အား|a: ဖြင့်|hpjin. မ|ma-ခက်|khe' ခဲ|ge: သော|tho: ပါ|ba
+ AEC + IPA + Align	ရေ|jei စီး|zi: ကြောင်း|gyaun: မှ|mha. ထွက်|htwe' လာ|la သည်|thi နှင့်|ne. ရေ|jei ကူး|gu: ခြင်း|gyin: သည်|dhe ပုံ|boun မှန်|mhan အား|a: ဖြင့်|hpjin. မ|ma-ခက်|khe' ခဲ|ge: တော့|do. ပါ|ba

TABLE VII: Error Comparison of Different AEC approaches for Whisper Tiny Model (Errors are colored in Red).

- The **plain AEC** model corrects most key tokens but may introduce minor replacements (e.g., "မှ|mha" → "မှာ|mha").
- **AEC + IPA** improves phonetic alignment, producing nearly identical output to the reference.
- **AEC + Align** enhances word order and structure but can still insert minor errors like "ပြီ|pji".

7. Future Work

- Explore **smarter augmentation strategies** aligned with test data conditions.
- Extend AEC to **multimodal settings** (e.g., audio-text correction with visual cues).
- **Integrate large language models (LLMs)** to enhance contextual correction.
- Publicly **release fine-tuned Whisper and AEC models**, OpenNMT configurations, and the **parallel Burmese AEC dataset** to support further research in low-resource ASR.

8. Conclusion

- This work investigated **ASR Error Correction (AEC)** for **low-resource Burmese** using various linguistic and alignment features.
- All AEC-based methods consistently **outperformed baseline ASR outputs**, confirming the effectiveness of error correction.
- **Alignment features** delivered the strongest and most consistent improvements across models.
- **IPA (phonetic) features** provided mixed results helpful in smaller models, but less effective for large backbones.
- **Data augmentation** improved robustness in smaller models but sometimes caused a **distribution mismatch** in stronger models.
- Results highlight the **robustness of AEC** and the importance of **careful feature selection** for low-resource speech processing.